Evaluation runs: abd (abd-register-civl)

generated 2026-11-08 from brain-data 38584f536 2026-11-08 20:45:11 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	3	12480	conforms	0.4	2	1 h, 250 GB	3	6	brain03, 2026-11-01
3	3		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-01

stateless: pair **read-no-writeback**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	3	544	conforms	0.1	2	1 h, 250 GB	3	6	brain03, 2026-11-01
3	3	960768	violation	6.7	4	1 h, 250 GB	20739	62214	brain03, 2026-11-01
4			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-01

Not run (stateless). skipped: a smaller size had reached its limit: 4 correct bag, 5 read-no-writeback bag, 5 correct bag, 6 correct bag, 6 read-no-writeback bag, 7 read-no-writeback bag, 7 correct bag, 8 read-no-writeback bag, 8 correct bag, 9 correct bag, 9 read-no-writeback bag.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	3	12480	conforms	0.6	2	1 h, 250 GB	brain02, 2026-11-01
3			🕓 (1 h)			1 h, 250 GB	brain02, 2026-11-01

stateful: pair **read-no-writeback**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	3	544	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-01
3	3	960768	violation	6.3	4	1 h, 250 GB	brain02, 2026-11-01
4			🕓 (1 h)			1 h, 250 GB	brain02, 2026-11-01

Not run (stateful). skipped: a smaller size had reached its limit: 4 correct bag, 5 read-no-writeback bag, 5 correct bag, 6 correct bag, 6 read-no-writeback bag, 7 read-no-writeback bag, 7 correct bag, 8 read-no-writeback bag, 8 correct bag, 9 correct bag, 9 read-no-writeback bag.

offline

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	3	12480	conforms	0.6	8	10 min, 16 GB	0.1	0.6	0.0	in-memory	brain05, 2026-11-02

offline: pair **read-no-writeback**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	3	544	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3	3	960768	violation	6.9	10	10 min, 16 GB	0.0	6.9	0.0	in-memory	brain05, 2026-11-02

Not run (offline). skipped: a smaller size had reached its limit: 3 bag correct, 4 read-no-writeback bag, 4 correct bag, 5 read-no-writeback bag, 5 correct bag, 6 correct bag, 6 read-no-writeback bag, 7 read-no-writeback bag, 7 correct bag, 8 read-no-writeback bag, 8 correct bag, 9 correct bag, 9 read-no-writeback bag.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	3	12480	conforms	3.3	164	10 min, 16 GB	3.3	0.6	0.0	in-memory	brain01, 2026-11-02
3	3	5247327744	conforms	81614.0	16495	48 h, 200 GB	4.2	81614.0	0.0	merge	volta14, 2026-11-08

cross offline, Rust implementation against the Java specification: pair **read-no-writeback**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	3	544	conforms	3.3	164	10 min, 16 GB	3.3	0.1	0.0	in-memory	brain01, 2026-11-02
3	3	960768	violation	6.9	163	10 min, 16 GB	2.9	6.9	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 4 bag read-no-writeback.

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	3	12480	conforms	79.6	1870	1 h, 250 GB		brain05, 2026-11-02
3	3		🕓 (1 h)			1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **read-no-writeback**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	3	544	conforms	3.6	559	1 h, 250 GB		brain05, 2026-11-02
3	3		🕓 (1 h)			1 h, 250 GB		brain05, 2026-11-02

Not run (cross online, Java implementation (JMC) against the Rust specification export). planned but never scheduled (grid points beyond the ladder): 2 bag correct, 2 bag read-no-writeback, 3 bag correct, 3 bag read-no-writeback.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	3	12480	conforms	4.0	160	1 h, 250 GB		brain01, 2026-11-02
3	3		(1 h)			1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **read-no-writeback**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	3	544	conforms	3.4	160	1 h, 250 GB		brain01, 2026-11-02
3	3	960768	violation	10.9	157	1 h, 250 GB	309	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12480	12480	conforms	64.9	5612	10 min, 16 GB	64.9	0.6	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair **read-no-writeback**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12480	544	conforms*	64.9	5612	10 min, 16 GB	64.9	0.1	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair **read-no-writeback**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	544	544	conforms	5.6	279	10 min, 16 GB	5.6	0.1	0.0	in-memory	brain01, 2026-11-02
3	960768	960768	conforms	2201.8	30799	10 min, 16 GB	2201.8	6.9	0.0	in-memory	brain01, 2026-11-02

Not run (twins offline, Rust implementation against its Java twin). skipped: a smaller size had reached its limit: 3 bag correct, 4 bag correct, 4 bag read-no-writeback.
refused (an input missing at run time): 3 bag read-no-writeback.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12480	12480	conforms	64.9	5612	10 min, 16 GB	0.6	64.9	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **read-no-writeback**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12480	544	conforms*	5.6	279	10 min, 16 GB	0.6	5.6	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **read-no-writeback**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	544	544	conforms	5.6	279	10 min, 16 GB	0.1	5.6	0.0	in-memory	brain01, 2026-11-02
3	960768	960768	conforms	2201.8	30799	10 min, 16 GB	6.9	2201.8	0.0	in-memory	brain01, 2026-11-02

Not run (twins offline, Java implementation against its Rust twin). skipped: a smaller size had reached its limit: 3 bag correct, 4 bag correct, 4 bag read-no-writeback.
refused (an input missing at run time): 3 bag read-no-writeback.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12480	12480	conforms	117.1	9146	10 min, 16 GB	117.0	1.1	0.1	in-memory	brain02, 2026-11-02

context-free twins offline, Rust implementation against the context-free Java port: pair **read-no-writeback**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12480	544	conforms*	117.0	9144	10 min, 16 GB	117.0	0.2	0.0	in-memory	brain02, 2026-11-02

context-free twins offline, Rust implementation against the context-free Java port: pair **read-no-writeback**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	544	544	conforms	10.3	1702	10 min, 16 GB	10.3	0.2	0.0	in-memory	brain02, 2026-11-02

Not run (context-free twins offline, Rust implementation against the context-free Java port). skipped: a smaller size had reached its limit: 3 bag correct.
refused (an input missing at run time): 3 bag read-no-writeback.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12480	12480	conforms	117.0	9146	10 min, 16 GB	1.1	117.0	0.0	in-memory	brain02, 2026-11-02

context-free twins offline, Java port against the Rust implementation: pair **read-no-writeback**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12480	544	conforms*	10.3	1704	10 min, 16 GB	1.1	10.3	0.0	in-memory	brain02, 2026-11-02

context-free twins offline, Java port against the Rust implementation: pair **read-no-writeback**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	544	544	conforms	10.3	1702	10 min, 16 GB	0.2	10.3	0.0	in-memory	brain02, 2026-11-02

context-free twins offline, Java port against the Rust implementation: pair **read-no-writeback**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
3	960768	960768		conforms*	1644.9	17768	48 h, 250 GB	13.4	1644.9	0.0	merge	brain02, 2026-11-08

Not run (context-free twins offline, Java port against the Rust implementation). skipped: a smaller size had reached its limit: 3 bag read-no-writeback, 3 bag correct.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	12480	12480	conforms	117.2	9686	1 h, 250 GB		brain02, 2026-11-02

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **read-no-writeback**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	12480	544	conforms*	12.2	2359	1 h, 250 GB		brain02, 2026-11-02

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **read-no-writeback**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	544	544	conforms	11.1	2218	1 h, 250 GB		brain02, 2026-11-02
3	960768	960768	conforms	3195.7	36433	1 h, 250 GB		brain02, 2026-11-02

Not run (context-free twins online, Java port (JMC) against the Rust implementation's export). skipped: a smaller size had reached its limit: 3 bag read-no-writeback, 3 bag correct.

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	12480	12480	conforms	118.1	9140	1 h, 250 GB		brain02, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **read-no-writeback**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	12480	544	conforms*	117.1	9140	1 h, 250 GB		brain02, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **read-no-writeback**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	544	544	conforms	10.5	1697	1 h, 250 GB		brain02, 2026-11-02

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).
skipped: a smaller size had reached its limit: 3 bag read-no-writeback, 3 bag correct.